OLYMPIAD · EVERYDAY MATHEMATICS · OLYMPIAD PREPARATION · SHORT NOTES

Chapter 17: Area, Perimeter and Volume in Real Situations

Topic-wise short notes with solved examples

Fencing, tiling, painting, paths and tanks: using area, perimeter and volume formulas for everyday problems.

1 Which formula to use

Shape	Perimeter	Area
Rectangle, length l and breadth b	$2(l+b)$	$l \times b$
Square of side a	$4a$	a^2
Triangle	sum of sides	$\frac{1}{2} \times \text{base} \times \text{height}$
Circle of radius r	$2\pi r$	πr^2
Trapezium	sum of sides	$\frac{1}{2}(a+b) \times h$

- **Perimeter** is a length. Use it for **fencing**, a border, a rope or a wire.
- **Area** is the surface. Use it for **tiling**, carpets, painting, grass and a path.
- **Volume** is the space inside. Use it for tanks, boxes and filling with water.
- Take $\pi = \frac{22}{7}$ when the radius is a multiple of 7.
- Change all lengths to one unit first. Remember $1\ \text{m} = 100\ \text{cm}$.

Example 1.1 A rectangular field is 40 m long and 25 m wide. Find the cost of fencing it at Rs 12 per metre.

SOLUTION

Perimeter $= 2(40+25) = 130$ m.

Cost $= 130 \times 12$.

Rs 1560

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Example 1.2 A circular park has radius 14 m. Find its circumference and its area.

SOLUTION

Circumference $= 2 \times \frac{22}{7} \times 14 = 88$ m.

Area $= \frac{22}{7} \times 14 \times 14 = 616$ square metres.

88 m and 616 square metres

2 Tiles, carpets and walls

- Number of tiles $= \frac{\text{area of the floor}}{\text{area of one tile}}$. Use the same units for both.
- Cost $=$ area $\times$ rate per unit area.
- The four walls of a room of length l, breadth b and height h have area $2h(l + b)$. Subtract doors and windows when they are not painted.

Example 2.1 A room 6 m by 4 m is covered with square tiles of side 50 cm. How many tiles are needed?

SOLUTION

Floor $= 6 \times 4 = 24$ square metres. Tile $= 0.5 \times 0.5 = 0.25$ square metres.

Tiles $= \frac{24}{0.25}$.

96 tiles

Example 2.2 A room is 5 m long, 4 m wide and 3 m high. Find the area of the four walls.

SOLUTION

$2 \times 3 \times (5 + 4) = 54$.

54 square metres

3 Paths and tanks

- To find the area of a **path** of width w around a rectangle, find the area of the bigger rectangle (outside) and subtract the area of the smaller one.
- The volume of a **cuboid** is $l \times b \times h$. A cube of side a has volume a^3.
- $1\ \text{m}^3 = 1000$ litres and $1000\ \text{cm}^3 = 1$ litre.

Example 3.1 A path 2 m wide runs around the outside of a garden 20 m by 15 m. Find the area of the path.

SOLUTION

Outer rectangle $= (20 + 4) \times (15 + 4) = 24 \times 19 = 456$.

Garden $= 20 \times 15 = 300$.

156 square metres

Example 3.2 A tank is 2 m long, 1.5 m wide and 1 m deep. How many litres of water can it hold?

SOLUTION

Volume $= 2 \times 1.5 \times 1 = 3$ cubic metres.

3000 litres

REMEMBER

Draw the shape and write the lengths on it. The most common mistake is to use area when the question needs perimeter, or the other way round.

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